

MUHAMMAD HANZALA WAHID

AI Engineer — LLMs, AI & Machine Learning — Python

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SUMMARY

AI Engineer experienced in building production-oriented AI systems across LLM applications, voice AI, RAG, NLP, computer vision, document intelligence, and automation. Strong Python/FastAPI background with hands-on experience integrating LLMs, AI agents, APIs, databases, cloud workflows, and asynchronous systems from data processing through deployment.

TECHNICAL SKILLS

- **Programming:** Python, C, SQL, R (Basic)
- **AI/ML:** Machine Learning, Deep Learning, Feature Engineering, Classification, Ranking, Model Evaluation, ANN, CNN, TensorFlow, Keras, PyTorch, YOLOv8
- **LLM/GenAI:** LLM Applications, RAG, AI Agents, Multi-Agent Systems, Prompt Engineering, Embeddings, Groq, OpenAI
- **NLP/Document AI:** NLP, OCR, Document Extraction, Classification, Information Extraction, spaCy, Sentence Transformers, PyMuPDF, pdfplumber
- **Voice AI:** LiveKit Agents/SIP, Retell AI, Deepgram STT/TTS, Groq, OpenAI, Silero VAD
- **Computer Vision:** YOLOv8, OpenCV, Object Detection, Image Classification, Visual Compliance
- **Backend/Systems:** FastAPI, Django, Flask, REST APIs, Celery, Redis, PostgreSQL, MySQL, MongoDB, SQLite
- **Automation/Cloud:** n8n, Make, Zapier, Playwright, Google APIs, Azure Blob Storage, Docker, Git, GitHub, Postman

EXPERIENCE

AI Engineer

Apr 2026 – Present

Technaptix

- Build AI, automation, and backend systems across voice AI, LLM applications, document intelligence, computer vision, sales automation, lead generation, and data workflows.
- Develop Python/FastAPI services, AI agents, LLM integrations, automation pipelines, and cloud data workflows; optimize prompts, tokens, model selection, VAD, latency, reliability, and API costs.
- Integrate production AI systems with external APIs, SIP/telephony providers, databases, scheduling platforms, and workflow automation tools to deliver end-to-end business solutions.
- Contribute across the full AI engineering lifecycle, from prototyping and model/API evaluation to backend implementation, testing, optimization, and deployment.

AI Engineer (Project-Based)

Mar 2025 – Mar 2026

Rasinnovatech

- Developed AI and data-processing solutions across OCR, NLP, document intelligence, recruitment, lead generation, automation, and business workflows.
- Built AI Resume Analyzer, OneLinker CRM/lead workflows, Google Calendar automation, OCR/NLP pipelines, and data systems including a 1,750+ column Perff.com dataset and PakWheels automotive analysis.

SELECTED PROFESSIONAL PROJECTS – TECHNAPTIX

Outbound AI Voice Agent — Python, LiveKit SIP, Retell AI, Telnyx, Deepgram, Groq, OpenAI, n8n

- Built outbound AI calling workflows for lead qualification, sales conversations, transcripts, email, and Cal.com booking; optimized VAD/model usage and reduced estimated call cost to approximately PKR 15/call.

AI Voice Receptionist — Python, LiveKit Agents, Deepgram, Groq, ChromaDB, OpenAI

- Developed inbound voice AI with knowledge-base retrieval and tool-based business actions, achieving approximately 1.1–1.2s response latency and confirming bookings only after successful tool execution.

Pharmaceutical Advertising Compliance — Python, FastAPI, OCR, OpenAI Vision, React, TypeScript

- Built automated PDF/PPT/DOCX compliance checking across approximately 160 validation endpoints with parallel processing and pass/fail JSON/HTML reporting for exception-based review.

Sales Data Automation — Python, Playwright, openpyxl, Azure Blob Storage

- Automated daily month-to-date sales processing, transformation, Excel cleanup, dataset generation, and Azure upload, eliminating repetitive manual spreadsheet processing.

Industrial Safety Vision — Python, YOLOv8, OpenCV, PyTorch, FastAPI

- Developed vision pipelines for person, PPE, restricted-zone, and worker-idle detection, converting camera observations into structured safety events.

Lead Generation & Contact Intelligence — Python, FastAPI, Groq

- Built a scraping and enrichment POC identifying C-level/VP contacts, contact information, and scored outreach lists.

ERP/SAP Data Scraper — Python, HTTPX, BeautifulSoup, SQLite

- Developed a resumable SAP table-catalog scraper with local searchable storage and recovery from interrupted jobs.

Email Automation Platform — Python, FastAPI, SMTP, React

- Built personalized email automation with contact/template management, bulk sending, and campaign-level send logging through the company mailbox.

PERSONAL & ACADEMIC PROJECTS

AI-Powered Explainable Recruitment System – Final Year Project

- Built a FastAPI/Python recruitment platform using SQLAlchemy, spaCy, Sentence Transformers, PyMuPDF, and pdfplumber for CV parsing, job matching, ranking, anonymization, and explainable decisions.
- Implemented Top-5 candidate ranking, Normal/Fair evaluation modes, and fairness metrics including demographic parity difference, disparate impact ratio, and equal opportunity difference.
- Containerized the backend with Docker, PostgreSQL, Nginx, and JWT authentication.

AI-Powered Code Review Assistant

- Built an asynchronous AI code review platform using FastAPI, Django, Celery, Redis, Groq, and GitHub Pull Request integration to analyze code for bugs, performance, maintainability, security, and best-practice violations.
- Designed asynchronous review pipelines with Celery and Redis for non-blocking processing, automatically generating structured findings and actionable feedback directly on GitHub Pull Requests.

AI Resume Intelligence Platform

- Developed a FastAPI/PostgreSQL resume intelligence system with NLP, LLM-assisted evaluation, provider abstraction, automated document processing, and Streamlit UI.
- Implemented resume parsing, skill extraction, candidate profiling, and AI-driven analysis using configurable LLM providers and rule-based fallbacks.
- Designed modular backend services and database workflows for scalable document processing, evaluation, and integration with future web-based interfaces.

EDUCATION

BS Artificial Intelligence — Dawood University of Engineering and Technology, Karachi, Pakistan June 2026

Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Data Structures & Algorithms, Database Systems, Artificial Intelligence, Software Engineering

CERTIFICATIONS

Google AI Essentials — Coursera, 2026 | **Google Data Analytics Professional Certificate** — Coursera, 2026

ACHIEVEMENTS & LEADERSHIP

- Merit Scholarship recipient for highest CGPA in second year.
- Webmaster, IEEE Computer Society University Chapter; contributed to technical initiatives and student activities.
- Conducted AI and Python workshops; International IEEE Member; Google Developer Community Member.